

Featuring safety. At a primary level during the development stage precautions should be taken to make the instrument more robust and resilient, so that it can resist external impacts and the sensors can also be kept safe.

Protection caps (plastic or metal, depending on equipment and requirement) are also incorporated in calibrators to ensure safety during measurement. "Avoiding human interface with proper coating ensures that the body is rugged and reliable," explains Shah. "These equipment are now more rugged compared to before," adds Vishwa.

The challenges, still

With the employment of T&M equipment in a multitude of applications, various forms of calibrators (calibration equipment) have come to be in high demand. However, some challenges will have to be overcome in order to get the best in calibration.

Mobile calibrators. A trend developing for maintenance, repair and operation is the use of handheld equipment. In order to consolidate calibration in handheld equipment, "There is a need to have a devoted service interface and calibration software," says Shah.

A main calibration board is used for calibration of instruments, and its integration with the software as well as the interface can be difficult. Apart from that, there could be the possibility of experiencing other minor hindrances like, "Instrument holder availability during calibration or ambient conditions necessary and so on," adds Shah.



Fig. 2: Thermistor



Fig. 3: Pneumator



Fig. 4: Mini wind tunnel

And then there is the calibration certificate

It is very difficult to judge the performance of an instrument without a set of calibration results. A calibration certificate contains certain information to fulfil its purpose of supporting traceable measurements. This information listed in ISO Guide 25 can be divided into the following categories:

- Identification of the instrument and owner
- Identification of the calibrating laboratory
- Make and model of the instrument
- Previous measurements made by the equipment

It might feel redundant to quote the obvious, but putting down the facts together in simple terms is the easiest way to go.

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Equipment versus labs

The prerequisite for stable and optimum conditions calls for maintaining the equipment under very specific environments. Calibration labs come as a suitable solution to this. Necessary ambient conditions for calibrators that can be maintained only in labs ensure accuracy and longer life of the instrument. "Soaking period required by instruments before calibration, devoted apparatus and equipment required for calibration can only be found in calibration labs," explains Shah.

Another major advantage of favouring calibration labs is the lab certification that is issued for instruments. Shah says, "This is a priority for all internal and external audits."

Krishna explains the

requirement of a full pre-adjustment report on instrument performance as the need to know precisely where and how far out of specification it was before calibration. "This is a growing requirement for ISO 9000 compliance," he adds.

However, when instruments cannot be taken out to calibration labs, because of certain limitations, the calibrators can be of a huge help. "We do not have to carry measuring instruments to calibration labs and can calibrate these using calibrators," says Shah.

The slow wave of calibration

Overall, calibration equipment have continued to improve in performance to ensure that these are capable of calibrating test equipment. We also see that the trend of connected devices and test equipment has made its way to calibrators as well.

Additionally, if a piece of equipment is going to govern the quality of function of other devices, it has to be of optimum quality, as, Krishna adds, "measured values are the basis of decisions." **EFY**